

Marginal Matching Is Not Enough: Conditional Style Leakage in Factorized Generative Models

Anonymous Authors

Anonymous Institution

anonymous@example.com

Abstract

Factorized generative models regularize a style latent z_s toward a fixed prior using a marginal statistic, $q(z_s) \approx \mathcal{N}(0, I)$, and treat this as a certificate that z_s is class-invariant. We show the certificate is invalid: matching the marginal places no constraint on the class conditionals $q(z_s | y=k)$, so z_s can be exactly Gaussian in aggregate while remaining maximally informative about the class label y . We call the resulting failure *conditional style leakage* and give a diagnostic for it based on inter-class style separation, linear probing, and HSIC. Across four unsupervised baselines (VAE, WAE-MMD, β -TCVAE, FactorVAE) and a case-study model, F-CS-WAE, every model with near-zero global MMD nonetheless allows a linear probe to recover y from z_s with 74–100% accuracy (chance 10%); on MNIST, F-CS-WAE reaches 99.15% clustering accuracy while class-conditional generation succeeds only 15% of the time under the sampling procedure that marginal matching is supposed to license. We test two remedies. A supervised per-class style MMD substantially reduces, but does not eliminate, the leakage, at a clustering cost that is dataset-dependent rather than a fixed price of the fix. A post-hoc class-conditional style prior fully repairs MNIST generation without retraining, but only partially repairs CIFAR-10, so a remedy validated on one dataset does not automatically generalize to another. The broader point is methodological: no divergence computed on $q(z_s)$ alone can certify $z_s \perp y$, and factorized generative models that report only marginal statistics have not actually verified the property they claim.

1 Introduction

Matching the aggregate style posterior to a Gaussian prior is a statement about a marginal distribution; class-invariant style is a statement about a family of conditional distributions. The two are not the same claim, and satisfying the first does not satisfy the second: $q(z_s)$ can be exactly $\mathcal{N}(0, I)$ while each $q(z_s | y=k)$ occupies a different region of latent space, so long as those regions average out to the right aggregate shape. A regularizer that only ever looks at $q(z_s)$ has no way to see this, and therefore no way to prevent it.

Where this bites. Consider a toy version of the standard factorized setup: a semantic variable z_c carries class identity, a style variable z_s is meant to carry everything else, and z_s is trained to satisfy $q(z_s) \approx \mathcal{N}(0, I)$. During training, the decoder only ever sees (z_c, z_s) pairs drawn from the same image, so if $q(z_s | y=k)$ happens to differ across classes—which nothing in the training objective prevents—the decoder is free to learn class-specific interactions between z_c and z_s . At generation time, a class prior supplies z_c for class k while z_s is drawn from the *marginal*, $\mathcal{N}(0, I)$, which averages over all classes. The resulting pair (z_c, z_s) is one the decoder never saw during training, and its output is unconstrained by anything the model learned. We call this failure *conditional style leakage*.

Why the usual metrics do not catch it. The failure is silent precisely because it is invisible to the metrics a practitioner would normally check. Reconstruction quality is measured on the posterior manifold – z_s inferred from a real image – where the class-specific interaction the decoder learned is exactly the one being exercised, so reconstruction looks fine. Global MMD is a marginal statistic by definition and, as above, is compatible with arbitrary conditional structure. Clustering accuracy depends only on the semantic variable z_c and says nothing about z_s at all. A model can therefore score well on every diagnostic a typical evaluation protocol reports and still fail completely at class-conditional generation, which is the one capability the factorization exists to provide.

A diagnostic, not just an example. Because the failure does not show up in standard metrics, detecting it requires measurements taken directly on the conditional structure of z_s : we use inter-class style separation Δ_{inter} , linear probing accuracy of y from z_s , and HSIC(z_s, y) (Section 3.4), alongside the marginal MMD that current practice already reports. A model with near-zero MMD but large Δ_{inter} , elevated linear-probe accuracy, or non-zero HSIC is leaking class information through a channel that was supposed to be class-invariant.

What we find. Applying this diagnostic across five model families on MNIST, every model with near-zero global MMD

still allows a linear probe to recover the class label from z_s with 74–100% accuracy against a 10% chance rate (Section 6.1); this includes β -TCVAE and FactorVAE, which are explicitly designed to penalize latent dependence. As a concentrated case study, we instantiate F-CS-WAE (Factorized Class-Structured Spherical Cauchy Wasserstein Auto-Encoder), a model that separates a hyperspherical semantic variable from a Euclidean style variable; without any conditional regularization it reaches 99.15% clustering accuracy on MNIST while class-conditional generation succeeds only 15% of the time. We then test two remedies – a supervised per-class style MMD applied at training time, and a post-hoc class-conditional style prior applied at sampling time – and report where each helps, where it does not fully fix the problem, and where a fix validated on MNIST fails to transfer to CIFAR-10 (Section 6.6).

Scope. We use F-CS-WAE to exhibit and repair the failure, not to argue it is a better model than the baselines it is compared against: the cross-model comparison in Section 6.1 does not control for encoder capacity or training budget, the CIFAR-10 baseline comparison in Section 5 is single-seed, and the proposed per-class style MMD is a supervised, label-guided regularizer, not an unsupervised disentanglement method. These scope limits are stated where they arise and collected in Section 7.1.

Contributions. Our contributions are fourfold, and each answers a question the previous one raises. First, we formalize the gap between marginal matching and conditional independence and prove it is unavoidable for any aggregate divergence, not a defect of a particular loss function (Proposition 1). This raises an obvious follow-up: is the gap only a theoretical curiosity, or does it bite in practice? Second, we build a lightweight diagnostic and measure it across five model families, showing the leakage is not only real but, for three of the five, invisible to a naive 2D projection of the latent (Section 6.1, Appendix E); we go beyond this correlational evidence with a causal latent-swap intervention and the full per-class/pairwise conditional-MMD structure behind Δ_{inter} (Section 6.3), showing the decoder does not merely have access to leaked class information but measurably conditions on it, and that this dependence itself shifts toward the semantic variable once the remedy is applied. Given that the failure is real, the natural question is whether it is fixable. Third, we propose per-class style MMD and show it substantially reduces, but does not eliminate, leakage, at a clustering cost that is dataset-dependent rather than uniform (Section 4); a six-point sweep over δ (Section 6.7) confirms leakage reduction is smooth in δ on both datasets, while showing clustering cost is noisier across δ than the original two-point comparison could reveal. Finally, because a fix validated on one dataset is not a fix validated in general, we test whether the same remedy transfers: the class-conditional sampling recipe that

is fully effective on MNIST only partially repairs CIFAR-10, a limitation invisible from the MNIST results alone (Section 6.6). We report this negative transfer result rather than omit it, since it constrains how far the proposed fix can be trusted to generalize.

2 Related Work

We organize prior work around a single question: does the method’s training or evaluation protocol certify conditional style invariance, $z_s \perp y$, or only a marginal or joint statistic that is compatible with $z_s \not\perp y$?

Variational and Wasserstein auto-encoders. VAEs [11] impose a per-example KL penalty toward a fixed Gaussian prior; WAEs [18] relax this to a divergence between the *aggregated* posterior and the prior, computed with kernel two-sample tests [4] or projected distances [13]. All of these regularize $q(z_s)$ or a per-example KL that, once aggregated over a labeled dataset, is again a statement about $q(z_s)$; none of them evaluates $q(z_s | y)$, so none certifies conditional invariance even when the marginal fit is good.

Disentanglement and total correlation. β -VAE [7] increases the KL weight to encourage disentangled factors; β -TCVAE [1] and FactorVAE [10] isolate and penalize the total correlation between latent dimensions directly. These methods get closer to our concern, since total correlation is a joint-distribution statistic rather than a purely marginal one, but the joint they penalize is over *latent dimensions*, not over the latent and the label. Table 2 shows both still leak: β -TCVAE and FactorVAE have among the lowest global MMD in our comparison, yet a linear probe recovers the class label from their style latent at 73.9% and 75.9% accuracy respectively, far above the 10% chance rate.

Conditional and semi-supervised generative models. Semi-supervised VAEs [12] and CVAE [17] condition generation on the label directly, which sidesteps needing z_c to encode class identity on its own, but neither checks whether the *residual* latent still carries label information after conditioning. StyleGAN [9] factorizes coarse structure from fine style via a mapping network in a class-unconditional regime, so the question does not arise there in the same form. Our setting requires both a class-conditional generator and a label-free clustering signal from the same model, which is what makes residual leakage in z_s consequential rather than incidental.

Deep generative clustering. DEC [20], IDEC [5], and VaDE [8] evaluate almost exclusively via ACC/NMI/ARI on the semantic or cluster-assignment variable. This protocol cannot see conditional style leakage by construction, since it never inspects z_s : our MNIST case study makes this concrete,

reaching 99.15% clustering accuracy (Section 6.1) on exactly the checkpoint whose class-conditional generation succeeds only 15% of the time (Section 6.2).

Hyperspherical latent variables. S-VAE [2] replaces the Gaussian posterior with a von Mises-Fisher distribution; Spherical Cauchy VAEs [16] give a heavier-tailed alternative with a Möbius-transform reparameterization. We use the latter for the semantic variable in our case-study model (Section 4.3). This choice of geometry is an implementation detail of the case study, not a claim about the leakage problem, which is defined independently of how z_c is parameterized.

3 The Conditional Style Leakage Problem

3.1 Setting and Notation

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ be a labeled dataset with K classes. A factorized model encodes each input into a semantic variable z_c and a style variable z_s . The style variable is regularized by

$$\mathcal{L}_{\text{style}} = D(q(z_s), \mathcal{N}(0, I)), \quad (1)$$

where D is any aggregate divergence and $q(z_s) = \mathbb{E}_y[q(z_s | y)]$. Class-conditional generation samples $z_s \sim \mathcal{N}(0, I)$ independently of z_c .

3.2 Formal Gap

Proposition 1 (Marginal matching does not certify conditional independence). *Let $z_s \sim \mathcal{N}(0, I)$ on $\mathbb{R}^{d_s}$, and let $\{A_k\}_{k=1}^K$ be any measurable partition of $\mathbb{R}^{d_s}$ with $P(z_s \in A_k) = 1/K$ for each k (such a partition always exists, e.g. via level sets of a continuous statistic of z_s). Define the label y as the deterministic function $y = k \iff z_s \in A_k$. Then:*

- (i) *the marginal is unchanged, $q(z_s) = \mathcal{N}(0, I)$ exactly, so $D(q(z_s), \mathcal{N}(0, I)) = 0$ for every divergence D ; and*
- (ii) *$I(z_s; y) = H(y) = \log K$, the maximum mutual information achievable between z_s and a K -ary label.*

Proof. (i) y is defined post hoc as a measurable function of z_s ; the distribution of z_s itself is never altered, so $q(z_s) = \mathcal{N}(0, I)$ holds exactly and every divergence to $\mathcal{N}(0, I)$, aggregate MMD included, is identically zero. (ii) Since y is a deterministic function of z_s , $H(y | z_s) = 0$, so $I(z_s; y) = H(y) - H(y | z_s) = H(y)$. By construction $P(y = k) = 1/K$ for all k , so $H(y) = \log K$, which is also the maximum possible entropy – and hence the maximum possible mutual information with any other variable – for a K -ary label. $\square$

Remark 1. *The construction is deliberately adversarial: $\{A_k\}$ need not resemble anything a trained decoder would produce, and $q(z_s | y=k)$ here is z_s restricted to A_k and renormalized, not a smooth family with class-specific means. That is the point – it shows that no divergence computed on*

$q(z_s)$ alone, MMD or otherwise, can rule out z_s being maximally class-informative, regardless of how the dependence is structured geometrically. Appendix D gives a smoother, empirically motivated construction with class-specific Gaussian conditionals that connects the same impossibility result to the finite-sample Δ_{inter} statistic we use as a diagnostic below; that construction is intuition for why Δ_{inter} is a reasonable statistic to track, not an alternative proof.

3.3 The Decoder Amplification Mechanism

During training the decoder receives (z_c, z_s) pairs from the same image; both encode y implicitly, z_s because $q(z_s | y=k)$ is not class-invariant. At generation time $z_c \sim p(z_c | y=k)$ encodes a specific class while $z_s \sim \mathcal{N}(0, I)$ averages over all classes, producing a cross-class pair never seen in training. Severity depends on how class-structured $q(z_s | y)$ becomes: visually uniform classes (MNIST digits) force more class information into z_s , amplifying the failure; high-diversity classes (CIFAR-10) leave more class-neutral variation in z_s . Section 6.5 tests this prediction directly.

3.4 Diagnostic Metrics

Global MMD $\text{MMD}^2(q(z_s), \mathcal{N}(0, I))$: near zero signals only that marginal matching succeeded.

Inter-class style separation $\Delta_{\text{inter}} = \frac{1}{\binom{K}{2}} \sum_{j < k} \|\bar{\mu}_s^{(j)} - \bar{\mu}_s^{(k)}\|_2$, $\bar{\mu}_s^{(k)} = \mathbb{E}[\mu_s | y=k]$: large values indicate class-dependent style means despite small global MMD. This is a finite-sample, mean-separation statistic; it is a convenient, easily computed symptom of conditional dependence, not what Proposition 1 is about. That proposition concerns statistical dependence in general (formalized as mutual information) and holds even when class-conditional means coincide, so long as higher-order structure differs.

Linear probing accuracy $\text{LP}(z_s \rightarrow y)$: accuracy of a linear classifier predicting y from z_s ; well above chance $1/K$ confirms recoverable class information.

HSIC(z_s, y) : kernel-based dependence measure [3]; non-zero indicates $z_s \not\perp y$ regardless of marginal shape.

A model with near-zero global MMD but large Δ_{inter} , elevated LP, or non-zero HSIC is exhibiting conditional style leakage.

4 Method

4.1 Problem Setting: Label-Guided Generative Clustering

Given labeled images $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ with K classes, we want (1) reconstruction that preserves visual content, (2) clustering that recovers class structure from semantic codes without using labels at evaluation time, and (3) class-conditional generation that produces class-consistent images from class priors.

4.2 General Remedy: Per-Class Style MMD

The remedy we propose is architecture-agnostic and applies to any factorized model with access to labels at training time. The standard, insufficient regularizer matches the marginal:

$$\mathcal{L}_{\text{style}} = \text{MMD}_{\text{RBF}}(\{z_s^i\}_{i=1}^b, \{\varepsilon_j\}_{j=1}^b), \quad \varepsilon_j \sim \mathcal{N}(0, I). \quad (2)$$

Our proposed fix instead matches each class conditional independently:

$$\mathcal{L}_{\text{style-cls}} = \frac{1}{K} \sum_{k=1}^K \text{MMD}_{\text{RBF}}(\{z_s^i : y_i=k\}, \{\varepsilon_j\}_{|y_i=k|}). \quad (3)$$

Where Eq. 2 has zero gradient incentive to reduce inter-class separation – the mixture can match $\mathcal{N}(0, I)$ while every conditional remains class-structured (Proposition 1) – Eq. 3 penalizes exactly that failure mode, by construction. Nothing about this term is specific to hyperspherical semantic variables, ResNet encoders, or any other choice made in the case-study instantiation of Section 4.3: it requires only a labeled batch and a style latent to regularize.

Scope of supervision. Because $\mathcal{L}_{\text{style-cls}}$ is computed per class, it requires labels at training time and is therefore a label-guided, supervised regularizer, not an unsupervised disentanglement method in the sense of β -TCVAE or FactorVAE. This is a different resource assumption than the four unsupervised baselines in Section 6.1, and we do not present the comparison between them as a like-for-like ablation of *how* to achieve independence, only as evidence that both unsupervised and supervised approaches leave measurable leakage. The *diagnostic* of Section 3.4, by contrast, requires only a labeled evaluation set and can be applied post hoc to any trained model regardless of how it was supervised, which is what makes it usable to audit the four unsupervised baselines in the first place.

4.3 Case-Study Instantiation: F-CS-WAE

To test the diagnostic and the remedy on a concrete model, we use F-CS-WAE (Factorized Class-Structured Spherical

Cauchy Wasserstein Auto-Encoder), which separates the latent into a hyperspherical semantic variable $z_c \in \mathbb{S}^{d_c-1}$, aligned to class-conditional Spherical Cauchy priors via a Möbius reparameterization [16], and a Euclidean style variable $z_s \in \mathbb{R}^{d_s}$ regularized by Eqs. 2–3. A shared ResNet-18 backbone feeds two heads producing $(\mu_c, \rho_c, \mu_s, \log \sigma_s^2)$; z_c and z_s are concatenated and decoded by a residual upsampling network. Full architectural detail is in Appendix A. The total training objective additionally includes a supervised semantic MMD toward the class prior ($\mathcal{L}_{\text{class}}$), an aggregated semantic MMD ($\mathcal{L}_{\text{agg}}$), and an auxiliary classification loss on μ_c ($\mathcal{L}_{\text{cls}}$) that prevents semantic collapse; these terms shape z_c for clustering and are not part of the style-leakage remedy itself. Coefficients are ramped over a four-phase, 300-epoch curriculum in which per-class style regularization begins in phase B, before the decoder has had a chance to learn class-specific (z_c, z_s) interactions (full schedule in Appendix C).

4.4 Inference and Generation

Clustering. K-means on deterministic μ_c , Hungarian-matched to compute ACC/NMI/ARI; labels are not used.

Naive prior sampling. $z_c \sim p(z_c | y=k)$, $z_s \sim \mathcal{N}(0, I)$. Reliable only when per-class style MMD is used.

Class-conditional style prior (post-hoc). Without per-class style MMD, estimate $\bar{\mu}_s^{(k)} = \mathbb{E}[\mu_s | y=k]$, $\bar{\sigma}_s^{2(k)} = \text{Var}[\mu_s | y=k]$ from encoded training data, then sample $z_s \sim \mathcal{N}(\bar{\mu}_s^{(k)}, \tau^2 \text{diag}(\bar{\sigma}_s^{2(k)})$ with $\tau = 0.25$. This is the second remedy we test, and it requires no retraining – only a forward pass over labeled training data to estimate $\bar{\mu}_s^{(k)}$ and $\bar{\sigma}_s^{2(k)}$.

5 Case Study Setup

Before using F-CS-WAE to study leakage, we check that it is a reasonably competent model on its own terms – otherwise the leakage findings below would be unsurprising, since a degenerate model could leak trivially. We report this briefly, as a sanity check rather than a headline result.

Datasets. CIFAR-10 [14] ($3 \times 32 \times 32$, $K=10$) is the primary clustering/generation benchmark. MNIST [15] and Fashion-MNIST [19] ($1 \times 32 \times 32$, $K=10$) are used for diagnostic analysis.

Table 1 shows clustering is stable across seeds (ACC std 0.52 points) and clearly separated from the baselines under the same protocol; we read this only as evidence that F-CS-WAE is not a degenerate model, not as a controlled superiority claim, since the baselines are single-seed and were not tuned to the same extent (Section 7.1). On MNIST, the single-latent CS-WAE variant (one Spherical Cauchy semantic latent, no style variable) reaches ACC 0.8735, NMI

Table 1: CIFAR-10 F-CS-WAE clustering/generation, three random seeds (all real), and comparison against single-seed baselines under the same unified lightweight-CNN protocol (Appendix B). Leakage diagnostics do not apply to these single-latent baselines.

Method	ACC	NMI	ARI	FID
F-CS-WAE (3-seed mean)	0.8087	0.6496	0.6325	83.04
std	0.0052	0.0079	0.0090	0.86
ResNetAE	0.2202	0.1017	0.0493	121.0
VAE	0.2220	0.0955	0.0511	180.0
WAE-MMD	0.2099	0.0979	0.0469	235.4
VaDE	0.1475	0.0427	0.0151	475.9

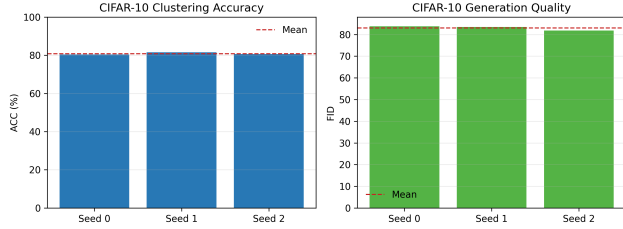


Figure 1: CIFAR-10 clustering and FID across three F-CS-WAE seeds. Variance is low.

0.8863, ARI 0.8411, FID 26.95 – included for context in Table 3 below, since it has no style latent and so cannot leak by definition. On Fashion-MNIST, F-CS-WAE (single seed) reaches ACC 0.9341, NMI 0.8672, ARI 0.8633, FID 72.37, sitting between MNIST and CIFAR-10 – a pattern we return to in Section 6.5.

Metrics used throughout. Clustering: ACC, NMI, ARI. Reconstruction: SSIM, PSNR, LPIPS [21]. Sample quality: FID [6] (10,000 generated vs. 10,000 real). Generation self-accuracy: fraction of generated images the trained auxiliary classifier assigns to the sampled class. Leakage: global MMD, Δ_{inter} , LP, HSIC.

6 Experiments

We organize the experiments by claim rather than by dataset: each subsection states what it tests, on which checkpoint, and what would falsify it.

6.1 Experiment 1: Marginal Metrics Hide Leakage

Claim tested: near-zero global MMD does not imply $z_s \perp y$, across model families, not just for F-CS-WAE.

We apply the diagnostic of Section 3.4 to four unsupervised baselines (VAE, WAE-MMD, β -TCVAE, FactorVAE) and F-CS-WAE on MNIST. The four baselines are purpose-built for this diagnostic only, with a lightweight 3-layer convolutional encoder trained for 100 epochs; they are *not* the same checkpoints as the like-named VAE/WAE-MMD entries used for CIFAR-10 clustering comparison in Section 5,

F-CS-WAE Class-Conditional Prior Samples: rows = class, columns = style samples

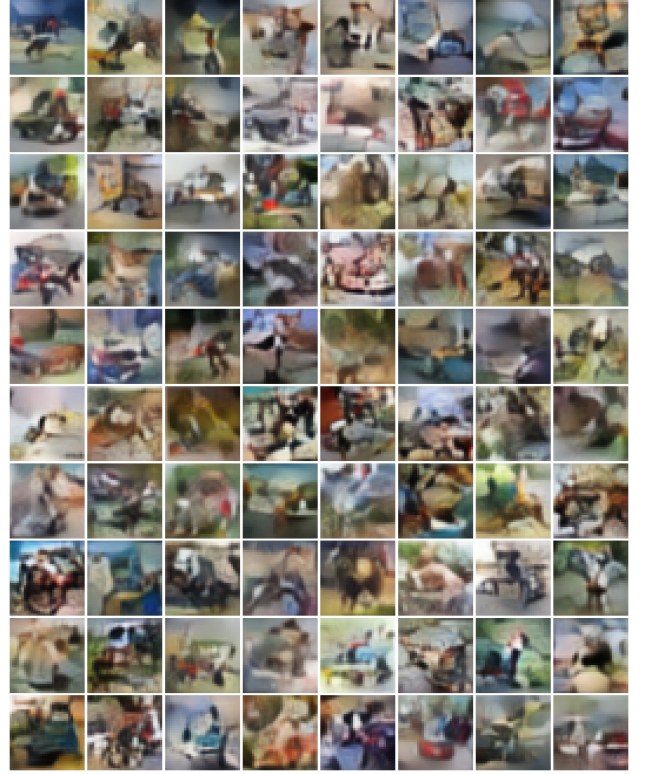


Figure 2: CIFAR-10 class-conditional prior samples (one row per class). FID 83 places these in the recognizable-but-blurry regime typical of L1/LPIPS-trained decoders without an adversarial loss, well short of photorealistic (Section 7).

which follow a different protocol (Appendix B). F-CS-WAE, by contrast, uses its full ResNet-18 encoder trained for 300 epochs (Section 4.3), so architecture and training budget are not matched across the five rows of Table 2 and the Δ_{inter} ranking between F-CS-WAE and the baselines should not be read as a controlled comparison. What the mismatch does not undermine is the *within-model* evidence: for every row individually, global MMD is decoupled from Δ_{inter} and LP regardless of that row’s encoder capacity or training length, which is the comparison the diagnostic actually depends on.

Result. All four baselines show LP far above chance (74–93%) despite near-zero global MMD. WAE-MMD has the largest global MMD among baselines (0.0576, 30–300 $\times$ the others), suggesting its own regularizer had not converged, and correspondingly the largest baseline Δ_{inter} (9.38) and LP (93%). β -TCVAE and FactorVAE, whose training objectives explicitly penalize latent dependence, show the smallest Δ_{inter} among the five models (2.30, 2.27) – but LP stays far above chance (74–76%): penalizing total correlation between latent *dimensions* reduces, but does not eliminate, de-

Table 2: Experiment 1. Conditional style leakage across five model families on MNIST. Despite near-zero global MMD in every case, inter-class style separation Δ_{inter} is large and linear probing (LP) recovers class information well above the 10% chance rate. The last row adds the remedy of Experiment 3. Self-ACC uses naive $z_s \sim \mathcal{N}(0, I)$ sampling.

Model	MMD ↓	Δ_{inter} ↓	LP ↓	Self-ACC ↑
VAE	0.0002	2.86	87.4%	N/A
WAE-MMD	0.0576	9.38	93.0%	N/A
β -TCVAE	0.0002	2.30	73.9%	N/A
FactorVAE	0.0009	2.27	75.9%	N/A
F-CS-WAE	0.0013	6.27	100.0%	0.15
F-CS-WAE+per-class	0.0015	1.46	45.6%	0.70

Table 3: Experiment 2. MNIST case study, single seed (seed 0). F-CS-WAE vs. the single-latent CS-WAE, which has no style latent to leak.

Method	ACC	NMI	ARI	FID	Δ_{inter}
CS-WAE	0.8735	0.8863	0.8411	26.95	—
F-CS-WAE	0.9915	0.9758	0.9813	73.99	6.27

pendence between the latent and the *label*, which is a different quantity. F-CS-WAE without the proposed remedy shows the most extreme leakage of the five ($\Delta_{\text{inter}} = 6.27$, LP= 100%), which Experiment 2 shows corresponds to a concrete generation failure, not just an elevated statistic.

Appendix E visualizes z_s for all six rows with t-SNE: the leakage is visually obvious only for WAE-MMD and F-CS-WAE (the two largest Δ_{inter} rows), while for VAE, β -TCVAE, and FactorVAE it is statistically real (LP 74–87%) but not visually apparent in a 2D projection – evidence that a plot is not a substitute for the quantitative diagnostic.

6.2 Experiment 2: Leakage Breaks Class-Conditional Generation

Claim tested: on a concrete checkpoint, leakage is not just a statistic – it corresponds to generation the auxiliary classifier and a human both judge as wrong.

Result. F-CS-WAE reaches 99.15% clustering accuracy and SSIM 0.983, yet class-conditional generation self-accuracy under naive sampling is only 15%: near-perfect scores on every standard metric coexist with a decoder that, when handed the cross-class (z_c, z_s) pairs that naive sampling produces, does not generate the requested class.

Figure 3 shows the failure is systematic rather than random: naive sampling does not spread errors uniformly across the nine wrong classes but collapses toward a small set of “attractor” digits (notably 8) regardless of which class was requested. Figure 4 makes the same point at the pixel level: naive-prior samples are legible digits, just frequently the *wrong* digit, which is consistent with the decoder resolving

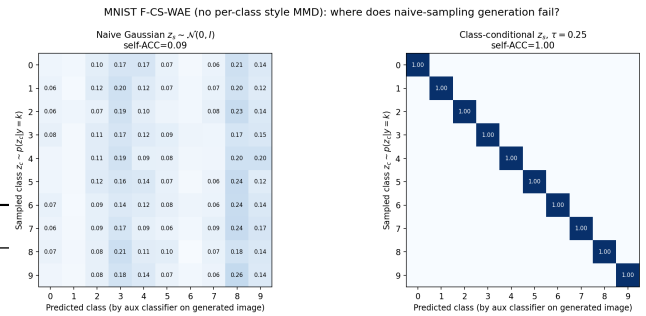


Figure 3: Confusion between the sampled class $y=k$ and the class the auxiliary classifier assigns to the generated image. Naive Gaussian sampling (left) does not fail randomly—probability mass collapses onto a handful of “attractor” digits (notably 8) regardless of the intended class. The class-conditional prior (right, $\tau=0.25$) is a clean diagonal.

an out-of-training-distribution (z_c, z_s) pair by falling back on whichever class its style input most resembles.

6.3 Direct Evidence: The Decoder Causally Uses Leaked Style

Claim tested: the linear probe of Section 3.4 shows z_s contains recoverable class information; it does not show the decoder *uses* that information to decide what to generate. We test this directly with two additional measurements on the same $\delta=0$ and $\delta=1$ F-CS-WAE checkpoints as Table 2.

The full conditional structure, not just its mean. Δ_{inter} summarizes class separation as a single mean-to-mean distance; it says nothing about how far each class conditional $q(z_s | y=k)$ itself sits from the prior it is supposed to match. We compute $\text{MMD}^2(q(z_s | y=k), \mathcal{N}(0, I))$ per class and pairwise $\text{MMD}^2(q(z_s | y=i), q(z_s | y=j))$ for all class pairs, on MNIST and CIFAR-10, at $\delta=0$ and $\delta=1$. At $\delta=0$, mean per-class MMD is $21\times$ the global MMD on MNIST (0.0186 vs. 0.00089) and $15\times$ on CIFAR-10 (0.0085 vs. 0.00056) – Figure 5 makes the point visually: every one of the ten per-class bars sits far above the global-MMD reference line the marginal regularizer actually optimizes. At $\delta=1$ the gap narrows but does not close (MNIST $4.4\times$; CIFAR-10 $5.3\times$), the same “reduces but does not eliminate” pattern Experiment 3 reports for Δ_{inter} and LP, now shown to hold for the full per-class MMD, not only its mean-separation proxy.

Latent swap: a causal intervention. For every ordered class pair (a, b) we take μ_c from a real test image of class a and μ_s from an independently drawn real test image of class b , decode $\text{Dec}(\mu_c, \mu_s)$, re-encode the result, and classify it with the same auxiliary classifier used for self-accuracy elsewhere. This is a stronger test than naive-prior sampling: it uses only real, in-distribution encoder outputs, so a failure to recover class a cannot be blamed on an out-of-distribution

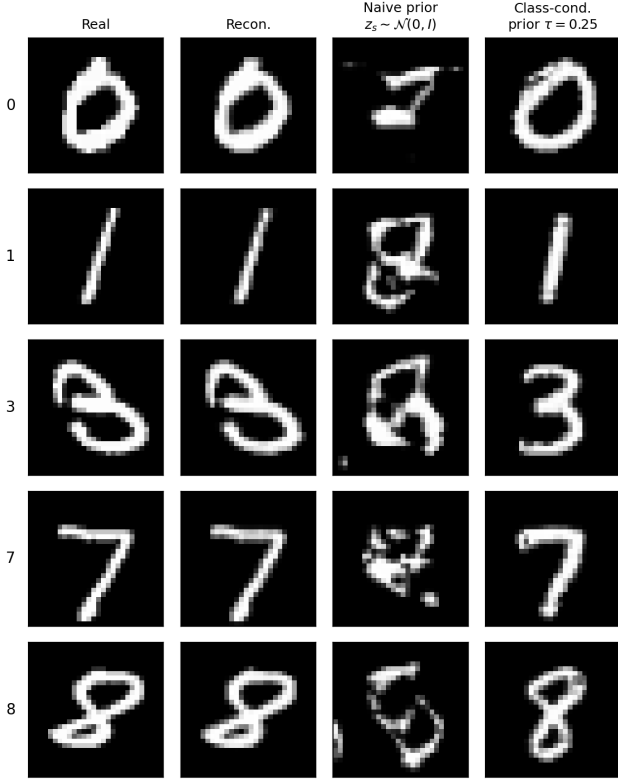


Figure 4: Real image, posterior reconstruction, naive-prior sample, and class-conditional-prior sample, for five representative digits. Reconstructions are visually indistinguishable from the real image (SSIM 0.983); naive-prior samples are legible digits, but frequently the *wrong* digit (e.g. row “0”→looks like a “3”; row “1”→looks like a “9”); the class-conditional prior restores the intended identity.

prior sample – it can only reflect what the decoder learned to do with a (z_c, z_s) pair drawn from two different real images. We report the diagonal-excluded means of $P(\text{pred}=a)$ (content-following) and $P(\text{pred}=b)$ (style-following).

Result. On MNIST at $\delta=0$, off-diagonal content-following is 0.1% and style-following is 99.2% (Figure 6, top): the decoder does not merely have access to leaked class information in z_s , it relies on that information almost exclusively to decide what to draw, discarding z_c almost entirely off-diagonal. Figure 7 shows why: every row (fixed z_c) produces visually identical output regardless of which class donated z_c ; only the column (the z_s donor) determines the digit. At $\delta=1$, content-following rises to 60.9% and style-following falls to 18.8% – a genuine causal change in what the decoder conditions on, not merely a reduction in a correlational statistic. On CIFAR-10 the same qualitative flip occurs but is weaker (content-following 4.2% → 20.0%, style-following 68.8% → 38.5%; heatmaps and grid in Appendix E, Figures 14–15), consistent with the cross-dataset asymmetry of Experiment 4 and Experiment 5: CIFAR-10’s higher intra-class diversity means the decoder was never as exclusively dependent on z_s for identity to begin with, so there is proportionally less for the

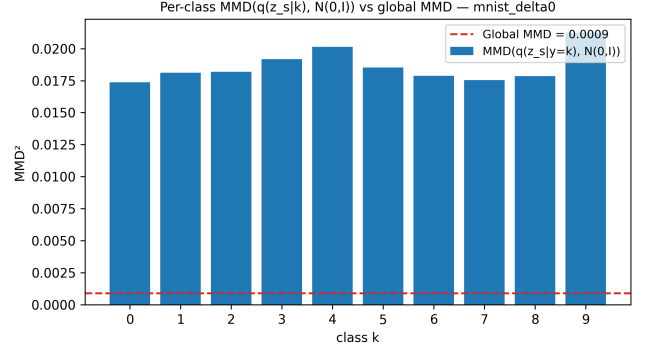


Figure 5: MNIST, $\delta=0$: per-class $\text{MMD}^2(q(z_s | y=k), \mathcal{N}(0, I))$ (bars) against the global MMD the marginal regularizer optimizes (dashed line). Every class sits far above the line the model was actually trained to minimize.

remedy to undo. The pairwise conditional-MMD structure behind Figure 5’s per-class bars, for both datasets, is likewise in Appendix E (Figure 13).

6.4 Experiment 3: Per-Class MMD Reduces but Does Not Eliminate Leakage

Claim tested: the remedy of Section 4.2 works in the direction intended, but does not drive leakage to zero, and reaching 100% self-accuracy requires combining it with the post-hoc prior.

Result. From Table 2, per-class style MMD cuts F-CS-WAE’s Δ_{inter} by 77% ($6.27 \rightarrow 1.46$) and LP by 54 points ($100\% \rightarrow 45.6\%$) – still far above the 10% chance rate, and higher than three of the four unsupervised baselines in the same table. Naive-sampling self-accuracy rises from 0.15 to 0.70, not 1.00: the remedy reduces how much class information z_s carries, but a linear probe can still recover the label well above chance, and naive sampling still fails on roughly 3 in 10 generations. Reaching 100% requires *combining* both remedies: applying the post-hoc class-conditional prior ($\tau=0.25$, Section 6.6) on top of the $\delta=1$ checkpoint, rather than relying on either alone, which we verified directly on this checkpoint (naive 0.70 → class-conditional 1.00, same protocol as Table 4). We read the two remedies as complementary rather than redundant: per-class style MMD reduces how much class information z_s carries, and the post-hoc prior corrects for what remains.

Why does combining remedies still work at $\delta=1$? The class-conditional sampling strategies above are themselves fit to per-class statistics of z_s , so it is natural to ask whether they remain distinguishable once $\delta=1$ has pulled every $q(z_s | y=k)$ toward $\mathcal{N}(0, I)$. We test this in latent space alone, with no decoding: an independent linear probe trained on real z_s (held-out accuracy 98.7% at $\delta=0$, only 32.3% at $\delta=1$, consistent with the LP drop above) is asked to classify syn-

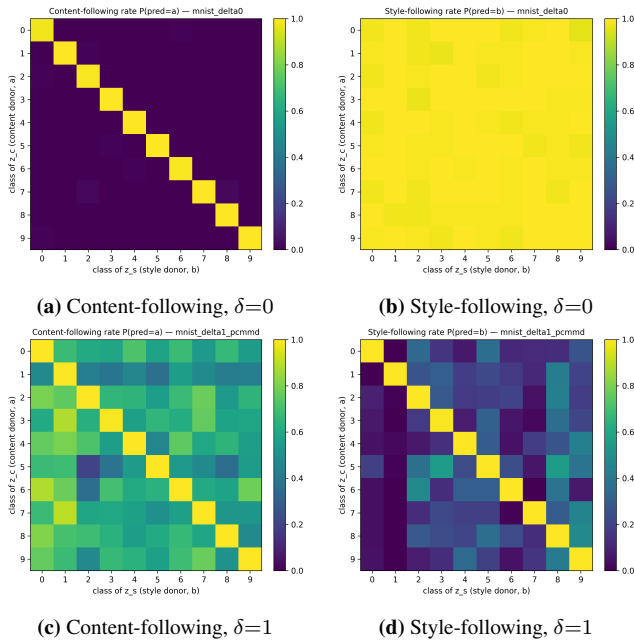


Figure 6: MNIST latent-swap intervention: rows are the class donating z_c (content), columns the class donating z_s (style). At $\delta=0$ (top), off-diagonal output is almost never the content class (mean 0.1%) and almost always the style class (mean 99.2%): the decoder has learned to read identity off z_s . At $\delta=1$ (bottom), this flips substantially (content 60.9%, style 18.8%).

thetic z_s draws from each sampling strategy. At $\delta=0$ every class-conditional strategy is $\geq 99.8\%$ self-consistent by this measure. At $\delta=1$, consistency collapses: the class-mean strategy falls to 80.0% and the genuinely-distributional strategies ($\tau=0.25$, $\tau=0.50$, empirical bank) fall to 15–28% – lower than at $\delta=0$, even though decoder self-accuracy under the same strategies reaches 100% (verified above). The resolution is Section 6.3’s causal intervention: at $\delta=1$ the decoder has shifted toward reading identity off z_c rather than z_s (content-following 0.1% $\rightarrow$ 60.9%), so the specific z_s value supplied at generation time matters less than it did at $\delta=0$. Combining remedies reaches 100% self-accuracy not because the sampled z_s becomes more class-distinguishable in latent space – by this measure it becomes *less* distinguishable – but because the decoder itself has become less sensitive to which z_s it receives.

6.5 Experiment 4: Leakage Tracks Intra-Class Diversity, Self-Accuracy Does Not

Claim tested: Section 3.3 predicts leakage severity should track intra-class visual diversity: MNIST (least diverse) $>$ Fashion-MNIST $>$ CIFAR-10 (most diverse). We test this on the $\delta=0$ F-CS-WAE checkpoint for all three datasets, using the same diagnostic throughout.

Result. Δ_{inter} and LP support the hypothesis cleanly: MNIST (6.27, 100.0%) $>$ Fashion-MNIST (4.86, 97.7%)

Latent swap: row = z_c class (content), col = z_s class (style) — mnist_delta0

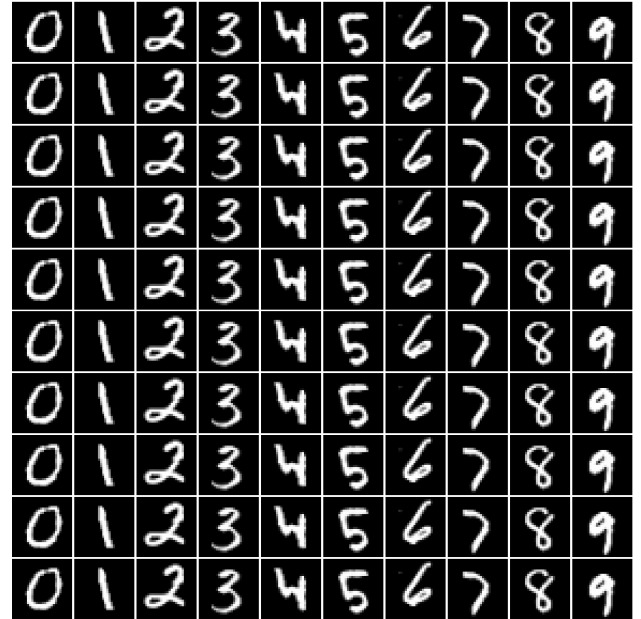


Figure 7: MNIST latent-swap grid, $\delta=0$: row = z_c donor class, column = z_s donor class. Every row is visually identical – identity is set entirely by the column.

$>$ CIFAR-10 (4.19, 89.9%). Naive-Gaussian generation self-accuracy does not: MNIST 0.15, Fashion-MNIST 0.17, CIFAR-10 0.12 – CIFAR-10 is lowest, not highest. The two families of metrics differ in what they are computed on, and that difference explains the disagreement rather than undermining either measurement. Δ_{inter} and LP operate directly on z_s and therefore isolate the dependency we care about. Self-accuracy is measured on the generated *image*, mediated by an auxiliary classifier, so it additionally inherits how confusable the real classes already are under blurry generation – a dataset-difficulty effect with nothing to do with conditional style leakage, and one that CIFAR-10 has on top of leakage. We see no principled reason to discard the metric that disagrees, and doing so would let us report a cleaner story than the data supports; the disagreement instead tells us that generation self-accuracy, on its own, is not a reliable cross-dataset proxy for leakage severity, even though it remains the right metric for asking whether a specific checkpoint’s generation is repaired (Section 6.6).

6.6 Experiment 5: Repair Does Not Transfer Uniformly Across Datasets

Claim tested: the class-conditional sampling recipe validated on MNIST (Experiment 2) transfers to CIFAR-10 with the same effectiveness.

On MNIST, every class-conditional strategy restores per-

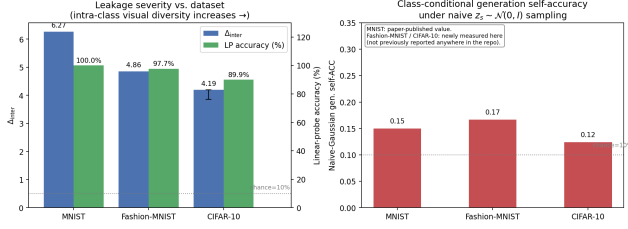


Figure 8: Left: Δ_{inter} and linear-probe accuracy across datasets, ordered by increasing intra-class visual diversity—both decrease monotonically, matching the hypothesis. Right: naive-Gaussian generation self-accuracy does *not* follow the same order (CIFAR-10 is lowest, not highest); see text.

Table 4: Experiment 5, MNIST. Generation under five z_s sampling strategies on the $\delta=0$ checkpoint; z_c always from the class prior.

Style sampling z_s	Self-ACC	Conf.	Div.
Global Gaussian	0.150	0.934	0.151
Class mean	1.000	1.000	0.132
Class diag., $\tau=0.50$	1.000	0.990	0.139
Class diag., $\tau=0.25$	1.000	1.000	0.135
Empirical style bank	1.000	1.000	0.182

fect self-accuracy (Table 4); the class-conditional diagonal Gaussian with $\tau=0.25$ is a true parametric sampler with perfect class consistency and better diversity than the class-mean point estimate.

Result. The $\tau=0.25$ recipe that is perfect on MNIST reaches only 68.1% on CIFAR-10 (Table 5), and this is not a sample-size artifact: the qualitative grid (Figure 9) shows why the number alone overstates the fix – samples within a class collapse toward a single low-diversity image, and several classes are not clearly identifiable by eye even where the classifier scores them correctly. Only the empirical style bank, which resamples real encoded style codes rather than fitting a per-class Gaussian, approaches MNIST-level self-accuracy (95.9%), suggesting CIFAR-10’s per-class style distribution is not well approximated by a unimodal Gaussian, unlike MNIST’s.

We treat this as a genuine limitation of the proposed remedy rather than omit it: the diagnostic of Experiment 1 and Experiment 4 generalizes across datasets more cleanly than the fix does, and a reader should not extrapolate the MNIST self-accuracy numbers to other datasets without checking.

6.7 Experiment 6: Dataset-Dependent Clustering-Leakage Trade-off

Claim tested: per-class style MMD’s cost to clustering accuracy is not a fixed price of the fix, but depends on how much class information the dataset had pushed into z_s in the first place.

Result. The original two-point comparison remains ac-

Table 5: Experiment 5, CIFAR-10. Same five sampling strategies, $\delta=0$ checkpoint. Unlike MNIST, the parametric class-conditional strategies plateau well below 100%.

Style sampling z_s	Self-ACC	Conf.	Div.
Global Gaussian	0.119	0.911	0.187
Class mean	0.700	0.996	0.079
Class diag., $\tau=0.50$	0.687	0.970	0.138
Class diag., $\tau=0.25$	0.681	0.975	0.101
Empirical style bank	0.959	0.988	0.241

Table 6: Effect of per-class style MMD at six δ values, both datasets, single seed per row. $\delta=0$ and $\delta=1$ rows reproduce Tables 2/3; the four intermediate values are new.

Dataset (δ)	ACC	Δ_{inter}	Naive FID
MNIST ($\delta=0$)	0.9915	6.27	73.99
MNIST ($\delta=0.03$)	0.9944	3.27	62.80
MNIST ($\delta=0.1$)	0.9898	2.12	52.61
MNIST ($\delta=0.3$)	0.9936	1.55	59.44
MNIST ($\delta=1$)	0.8592	1.46	43.36
MNIST ($\delta=3$)	0.9929	1.18	41.54
CIFAR-10 ($\delta=0$)	0.8149	4.19	83.59
CIFAR-10 ($\delta=0.03$)	0.8111	2.48	79.11
CIFAR-10 ($\delta=0.1$)	0.8100	2.03	77.50
CIFAR-10 ($\delta=0.3$)	0.7148	1.59	72.05
CIFAR-10 ($\delta=1$)	0.8000	1.28	74.42
CIFAR-10 ($\delta=3$)	0.7994	0.95	76.11

curate as far as it goes: on CIFAR-10, per-class style MMD is close to free at $\delta=1$ (81.49% $\rightarrow$ 80.00% ACC; the 3-seed mean without the regularizer is $80.87 \pm 0.52\%$, Table 1), Δ_{inter} drops 67%, and naive-sampling FID improves (83.59 $\rightarrow$ 74.42); on MNIST the leakage reduction is larger in relative terms (77%) but clustering accuracy drops substantially (99.15% $\rightarrow$ 85.92%).

The four additional δ values in Table 6 show this two-point picture, while accurate, is incomplete. Δ_{inter} and naive FID both decrease close to monotonically with δ on both datasets – the leakage-reduction mechanism behaves as expected across the whole range, not just at the two endpoints we originally measured. Clustering ACC does not move with it the same way: on MNIST, $\delta \in \{0.03, 0.1, 0.3, 3\}$ all sit at 99–99.4%, statistically indistinguishable from $\delta=0$, and $\delta=1$ is the *only* point with a large drop; on CIFAR-10, $\delta \in \{0, 0.03, 0.1, 1, 3\}$ all sit at 80–81%, and $\delta=0.3$ is the only point with a large drop (71.5%). We ran one seed per δ , so we cannot separate a genuine non-monotonic effect of δ on clustering from this particular run’s initialization at that one point; resolving that would require repeating each δ across multiple seeds, which is beyond this draft’s compute budget (Section 7.1). What the sweep does establish, which the two-point comparison could not, is directional: leakage reduction is a smooth function of δ on this evidence, while clustering accuracy is not, and neither dataset shows a clean monotonic

trade-off curve between the two at a single seed per point. The original interpretation – that the endpoint cost tracks how much a dataset’s model depended on the leak – remains consistent with the $\delta=0$ -vs- $\delta=1$ comparison it was based on, but should not be read as describing a smooth relationship across δ , which this sweep shows does not hold.

7 Discussion

Why CIFAR-10 is milder than MNIST, with a caveat.

CIFAR-10’s higher intra-class visual diversity weakens conditional style dependence, and $\Delta_{\text{inter}}/\text{LP}$ confirm this cleanly (Experiment 4). But Experiment 5 shows the practical remedy does not inherit this cleanly: the class-conditional Gaussian prior that perfectly fixes MNIST generation only partially fixes CIFAR-10 generation, and CIFAR-10 samples remain visibly blurry (FID ~ 83 – 84 , well above photorealistic ranges). We avoid describing CIFAR-10 naive-sampling output as “visually plausible”; at this FID the more accurate description is “recognizable but not sharp,” and under the $\tau=0.25$ class-conditional prior several classes are not reliably identifiable by eye despite scoring well on the auxiliary classifier (Figure 9).

Global vs. conditional regularization. Global aggregate matching is standard in WAEs and disentanglement models; our results show it is insufficient whenever the style variable must be class-invariant for generation. Per-class style MMD is a supervised analogue of total correlation penalties, targeting the single dependency $z_s \not\perp y$ directly rather than dependence among latent dimensions in general.

The reconstruction–generation gap. The MNIST checkpoint reaches SSIM 0.983 yet FID 73.99 under naive prior sampling (Figure 4). Reconstruction quality measures fidelity on the posterior manifold, not the prior; evaluation of generative clustering models should report both, since Section 6.1 shows the two can diverge arbitrarily.

7.1 Limitations

We list what this paper does *not* establish, rather than leave it implicit; several of these are previewed in the Scope paragraph of Section 1 and restated here in full. (1) Baseline comparisons (Table 1) are single-seed; we do not report multi-seed variance we have not measured, and the CIFAR-10 comparison is not a superiority claim for F-CS-WAE over the baselines, only evidence that it is not a degenerate model. (2) The clustering-leakage trade-off (Table 6, Figure 10) now spans six δ values per dataset rather than two, but each is a single training run; the non-monotonic ACC pattern we observe (Section 6.7) may reflect true δ -dependent instability at specific values or ordinary seed variance we cannot distinguish without repeating each δ across multiple seeds,

which we have not done. (3) The cross-model comparison (Experiment 1) does not control for encoder capacity or training budget between F-CS-WAE and the four baselines. (4) Per-class style MMD is a supervised, label-guided regularizer; we do not report a component-wise ablation of F-CS-WAE beyond the δ axis (e.g., removing the semantic variable, the class MMD term, or varying the prior geometry), since we do not have completed runs for those variants. (5) Generation self-accuracy is confounded by dataset-intrinsic classification difficulty and should not be used alone as a cross-dataset leakage-severity proxy (Experiment 4). (6) The class-conditional style prior’s failure to fully transfer to CIFAR-10 (Experiment 5) is measured on one checkpoint per dataset; we have not tested whether a richer conditional (e.g., a per-class mixture rather than a single Gaussian) closes the gap.

8 Conclusion

Conditional style leakage is a failure mode in factorized generative models: global aggregate regularization of a style variable does not prevent it from becoming class-dependent, because marginal matching is compatible with z_s being maximally informative about the class label, not just weakly correlated with it (Proposition 1). We measured this gap with a diagnostic across five model families and showed it is statistically real even when not visually obvious. A causal latent-swap intervention confirms the decoder does not merely have access to the leaked information but measurably conditions on it (off-diagonal content-following rises from 0.1% to 60.9% on MNIST once the remedy is applied). Per-class style MMD cuts inter-class style separation by 67–77% at the originally-measured $\delta=1$ and, across a fuller six-point δ sweep, reduces it smoothly on both datasets; the clustering cost is dataset-dependent and, at a single seed per δ , noisier than the original two-point comparison suggested – small across most of the sweep on both datasets, with one large outlier per dataset we cannot yet attribute to δ itself rather than seed variance. Per-class style MMD also raises MNIST naive-sampling self-accuracy from 15% to 70% (100% combined with the post-hoc style prior). Critically, the remedy that fully repairs MNIST generation only partially repairs CIFAR-10 generation, so fixing the diagnostic finding on one dataset does not automatically fix the underlying generative gap on another. Factorized generative models should verify class-invariance of style variables via conditional diagnostics, not marginal statistics alone, and should report where proposed remedies do and do not generalize rather than presenting a single-dataset result as general.

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A Architecture Details

The semantic head produces $(\tilde{\mu}_c, r_c)$, giving $\mu_c = \tilde{\mu}_c / \|\tilde{\mu}_c\|_2$ and concentration $\rho_c = \sigma(r_c)(1 - \varepsilon)$; the semantic latent is sampled by the Spherical Cauchy Möbius reparameterization $\eta_c \sim \text{Unif}(\mathbb{S}^{d_c-1})$, $z_c = T_{\mu_c, \rho_c}(\eta_c)$. The style head produces $(\mu_s, \log \sigma_s^2)$ and $z_s = \mu_s + \sigma_s \odot \varepsilon_s$, $\varepsilon_s \sim \mathcal{N}(0, I)$. The ResNet-18 backbone for 32×32 inputs replaces the first 7×7 convolution (stride 2) with a 3×3

convolution (stride 1, padding 1) and removes the max-pool layer, yielding a 4×4 spatial output with 512 channels; a shared FC layer maps $512 \times 4 \times 4 = 8,192$ to 256 dimensions with SiLU, from which separate linear heads produce the semantic and style parameters. The decoder concatenates $[z_c; z_s]$ ($d_c=64$, $d_s=128$ in all experiments) and maps it through three ResBlockUp stages (bilinear $2 \times$ up-sample, 1×1 projection, GroupNorm+SiLU+Conv residual block; $512 \rightarrow 256 \rightarrow 128 \rightarrow 64$ channels, $4 \times 4 \rightarrow 32 \times 32$), followed by a 3×3 output convolution and sigmoid. For each class k , F-CS-WAE maintains a center $m_k \in \mathbb{S}^{d_c-1}$ with prior $p(z_c | y=k) = \text{SCauchy}(m_k, \rho_p)$, $\rho_p=0.7$, updated by EMA from batch class-mean directions once per epoch, $m_k \leftarrow \text{normalize}(\tau m_k + (1 - \tau)\bar{\mu}_c^{(k)})$, $\tau=0.95$. The full training objective is

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \alpha(t)\mathcal{L}_{\text{class}} + \beta(t)\mathcal{L}_{\text{agg}} + \gamma(t)\mathcal{L}_{\text{style}} + \delta(t)\mathcal{L}_{\text{style-cls}} + \eta(t)\mathcal{L}_{\text{cls}}, \quad (4)$$

with $\mathcal{L}_{\text{rec}} = \lambda_1 \|x - \hat{x}\|_1 + \lambda_{\text{LPIPS}} \text{LPIPS}(x, \hat{x})$, $\mathcal{L}_{\text{class}} = \frac{1}{K} \sum_k \text{MMD}^2(Q_k, P_k)$ with $Q_k = \{z_c^i: y_i=k\}$, $P_k \sim p(z_c | y=k)$, $\mathcal{L}_{\text{agg}} = \text{MMD}^2(\{z_c^i\}, \{z_{c,r_j}^p\})$ against a random-class prior mixture, and $\mathcal{L}_{\text{cls}} = \text{CE}(h(\mu_c), y)$ a linear auxiliary classifier preventing semantic collapse.

B Training Hyperparameters

Table 7: Hyperparameters used in all F-CS-WAE experiments.

Parameter	Value
Optimizer	Adam, lr = 10^{-3}
LR schedule	StepLR, step 100, $\gamma = 0.5$
Batch size	128
Total epochs	300
Gradient clip (global norm)	1.0
Semantic dim d_c / style dim d_s	64 / 128
Prior concentration ρ_p	0.7
EMA momentum τ	0.95
$\lambda_1 / \lambda_{\text{LPIPS}}$	1.0 / 0.1
α_f / β_f (class / agg. MMD)	2.0 / 5.0
γ_f (style)	1.0
δ_f (per-class style, off/on)	0 / 1
η_f (classifier)	0.3
Style prior temperature τ_s	0.25

Baselines (ResNetAE, VAE, WAE-MMD, VaDE, CS-WAE) use a unified lightweight CNN protocol, 50 epochs, Adam lr 10^{-3} , StepLR(30, 0.5); the cross-model diagnostic baselines of Experiment 1 use a separate lightweight 3-layer CNN encoder, 100 epochs (see Section 6.1 for the resulting comparability caveat).

Table 8: Training phase boundaries and active loss terms; ramps are linear.

Phase	Epochs	Terms and ramp
A	0–49	$\mathcal{L}_{\text{rec}}$ only
B	50–99	$+\mathcal{L}_{\text{style}}(0 \rightarrow \gamma_f)$, $\mathcal{L}_{\text{style-cls}}(0 \rightarrow \delta_f)$, $\mathcal{L}_{\text{cls}}(0.1 \rightarrow 0.2)$
C	100–199	$+\mathcal{L}_{\text{class}}(0 \rightarrow \alpha_f/2)$, $\mathcal{L}_{\text{agg}}(0 \rightarrow \beta_f/2)$, $\mathcal{L}_{\text{cls}}(0.2 \rightarrow 0.3)$
D	200–299	$\mathcal{L}_{\text{class}}(\alpha_f/2 \rightarrow \alpha_f)$, $\mathcal{L}_{\text{agg}}(\beta_f/2 \rightarrow \beta_f)$

C Phase Schedule Details

Per-class style regularization ramps up in Phase B, before $\mathcal{L}_{\text{class}}$ and $\mathcal{L}_{\text{agg}}$ are introduced in Phase C, so that the decoder never has a window in which class-specific (z_c, z_s) interactions can form unchecked.

D Smooth Construction Motivating Δ_{inter}

Proposition 1 uses an adversarial partition construction to establish the impossibility result in its strongest, cleanest form. The following construction is not a stronger proof – it is a smoother, more empirically representative special case, included because it connects the same impossibility to the Δ_{inter} statistic used as a diagnostic throughout the paper.

Let $K=10$, $d_s=2$, and define class-conditional distributions $q(z_s | y=k) = \mathcal{N}(\mu_k, \Sigma_k)$ with $\mu_k = r(\cos(2\pi k/K), \sin(2\pi k/K))$ for radius $r>0$ and $\Sigma_k = I - \frac{r^2}{1+r^2} \hat{\mu}_k \hat{\mu}_k^\top$, $\hat{\mu}_k = \mu_k / \|\mu_k\|$ – a smooth family of Gaussian conditionals with class-specific means arranged on a circle, unlike the disjoint-support partition of the main-text proposition. The mixture mean is $\frac{1}{K} \sum_k \mu_k = 0$ and the mixture covariance is $I + \frac{r^2}{K} \sum_k \hat{\mu}_k \hat{\mu}_k^\top - \frac{r^2}{K} I$, which approaches I for this symmetric configuration. The inter-class mean separation is $\Delta_{\text{inter}} \approx r\sqrt{2}$, unbounded as $r \rightarrow \infty$, while the marginal stays $\approx \mathcal{N}(0, I)$: as r grows, this smooth, mean-separated construction drives Δ_{inter} to arbitrarily large values under an approximately matched marginal, illustrating in the same geometry as our empirical statistic why Δ_{inter} is worth tracking even though, unlike Proposition 1, this particular construction only becomes exact in the $r \rightarrow \infty$ limit.

E t-SNE of the Style Latent Across Model Families



Figure 9: CIFAR-10 naive (odd rows) vs. class-conditional $\tau=0.25$ (even rows) samples, four classes. Diversity within a row collapses under $\tau=0.25$, and object identity is often not clearly recognizable by eye even where the auxiliary classifier scores it correctly.

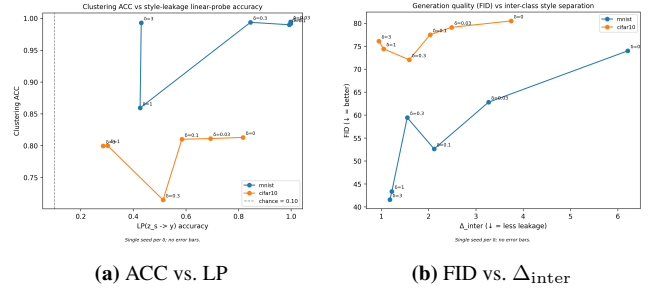


Figure 10: Six-point δ sweep ($\delta \in \{0, 0.03, 0.1, 0.3, 1, 3\}$), single seed per point, no error bars. Points are annotated with δ and connected in increasing- δ order per dataset; the connecting line is a visual aid, not a claim of a smooth underlying curve (see text). Leakage metrics (LP, Δ_{inter} , x -axis of both panels) move smoothly with δ ; clustering ACC does not.



Figure 11: t-SNE of the style latent z_s , colored by MNIST digit class, for all six rows of Table 2. Unlike a typical clustering figure, *color-separated clusters here mean worse* (class information leaking into z_s): separation is visually obvious only for WAE-MMD and F-CS-WAE (no per-class MMD)—the two rows with the *largest* Δ_{inter} in Table 2—and visibly reduced (improved) after adding per-class style MMD. For VAE, β -TCVAE, and FactorVAE the leakage is statistically real (LP 74–87%) but not visually apparent in this 2D projection. Referenced from Experiment 1 (Section 6.1).

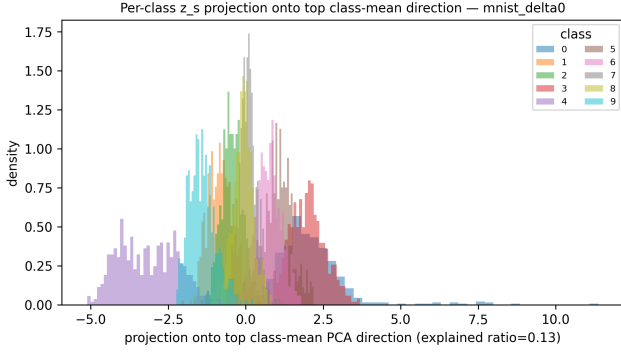


Figure 12: An alternative to the t-SNE view above, for F-CS-WAE at $\delta=0$: project z_s onto the top principal component of the $K \times d_s$ class-mean matrix (the same signal Δ_{inter} summarizes into one scalar) and plot per-class histograms. Some classes separate cleanly along this single direction even where a 2D t-SNE embedding does not make the separation visually obvious, illustrating that a fixed low-dimensional projection can either overstate or understate leakage depending on which directions it happens to preserve – another reason to prefer the quantitative diagnostics of Section 3.4 over any single plot.

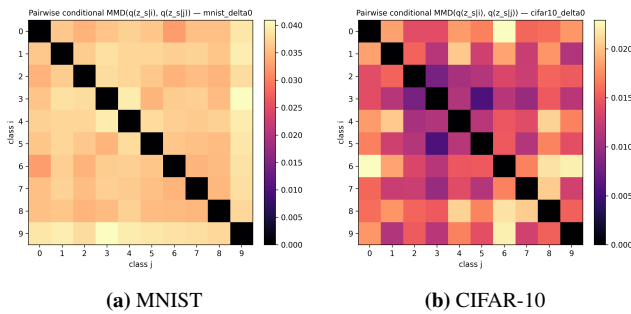


Figure 13: Pairwise $\text{MMD}^2(q(z_s | y=i), q(z_s | y=j))$ between all class pairs, F-CS-WAE at $\delta=0$. Off-diagonal structure shows which class pairs occupy the most visibly different regions of z_s ; both datasets show broad off-diagonal separation, not a few isolated confusable pairs, consistent with the per-class bar chart of Figure 5 showing every class contributes to the leakage rather than a small subset.

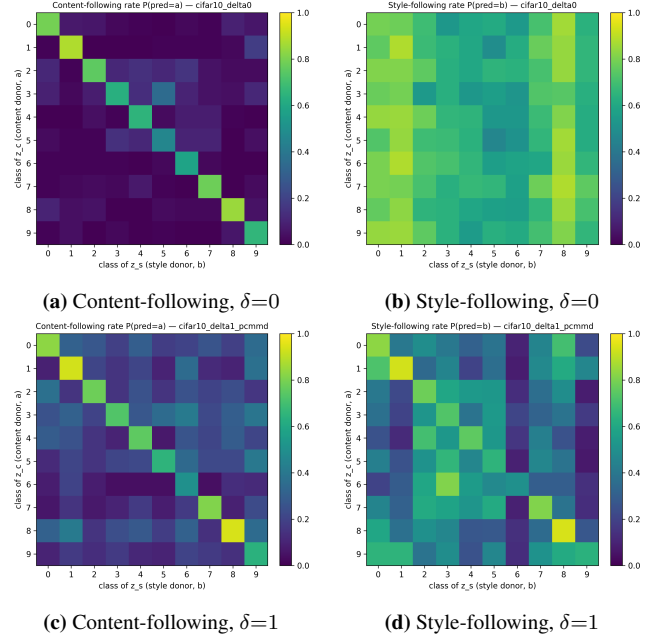


Figure 14: CIFAR-10 counterpart to Figure 6. The $\delta=0 \rightarrow \delta=1$ shift is the same direction as MNIST but weaker in magnitude (content-following 4.2% $\rightarrow$ 20.0%, style-following 68.8% $\rightarrow$ 38.5%), consistent with Section 6.3’s reading that CIFAR-10’s decoder was never as exclusively dependent on z_s for identity to begin with.

Latent swap: row = z_c class (content), col = z_s class (style) — cifar10_delta0

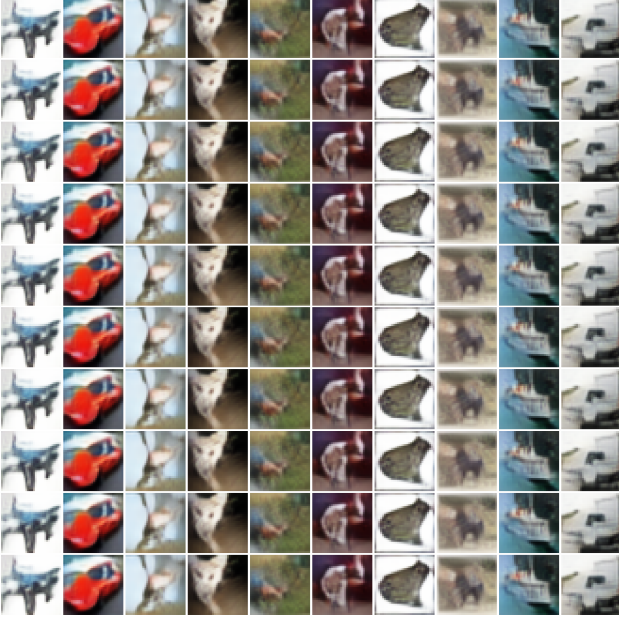


Figure 15: CIFAR-10 latent-swap grid, $\delta=0$ (cf. Figure 7 for MNIST). Rows are visually more similar to each other than in the MNIST grid but still show column-driven (style-driven) drift in object identity for several classes.

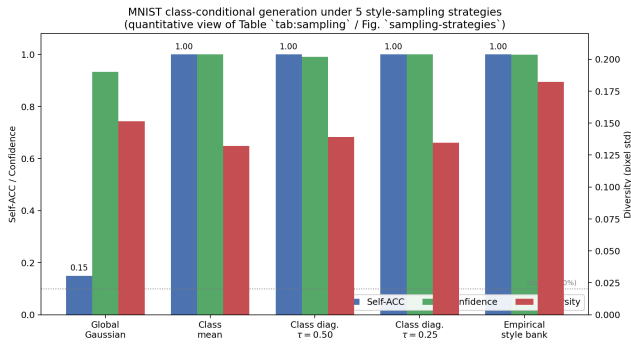


Figure 16: MNIST sampling strategies (Table 4) plotted directly: self-accuracy jumps from near-chance to 100% the moment z_s is conditioned on class in any way, while diversity (right axis) is lowest for the class-mean point estimate and highest for the empirical style bank.

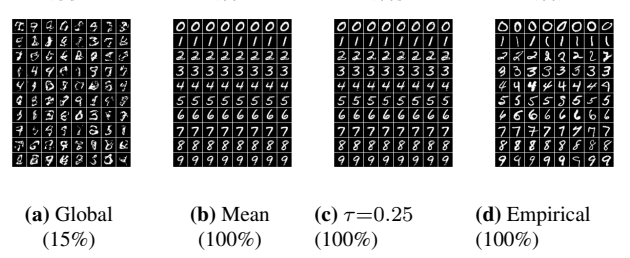


Figure 17: MNIST generation under representation-aware style sampling. Global Gaussian produces shapeless or wrong-class images; all class-conditional strategies restore correct digit identity.

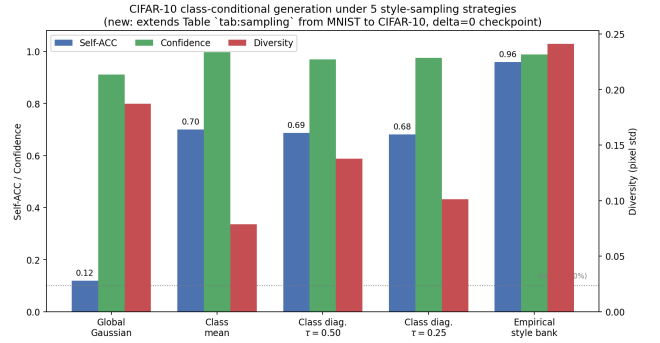


Figure 18: CIFAR-10 analogue of Figure 16 (Table 5). The class-conditional Gaussian strategies that reach 100% on MNIST plateau near 70% here; only resampling real encoded style codes (empirical bank) approaches MNIST-level self-accuracy.